

## Research Methodology and IPR (22MBA23)

### Module-1

#### Introduction to Business Research:

Meaning, types, process of research- management problem, defining the research problem, formulating the research Hypothesis, developing the research proposals, research design formulation, sampling design, planning and collecting the data for research, data analysis and interpretation. Research Application in business decisions, Ethical issues in business research. Features of a good research study.

### Module-2

#### Business Research Design:

Meaning, types and significance of research design, errors affecting research design.

**Exploratory Research:** Meaning, purpose, methods, Literature search, experience survey, focus groups and comprehensive case methods.

**Conclusive Research Design:** Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.

**Experimental Research Design:** Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

### Module-3

**Sampling:** Concepts, Types of Sampling, **Probability Sampling:** simple random sampling, systematic sampling, stratified random sampling, cluster sampling.

**Non Probability Sampling:** convenience sampling- judgmental sampling, snowball sampling, quota sampling, Errors in sampling.

### Module-4

**Data Collection:** Meaning, types, **Data collection methods:** Observations, survey and interview techniques, **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary Data Sources: advantages and disadvantages. **Measurement and Scaling Techniques:** Basic measurement Scales-Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale, **Multi-Dimensional Scaling:** Non comparative scaling techniques.

### Module-5

**Data Analysis and Report Writing:** Editing, Coding, Classification, Tabulation, Validation. Analysis and Interpretation, Report writing and presentation of results, Importance of report writing, types of research reports, Report structure, Guidelines for effective documentation.

### Module-6

#### Intellectual Property Rights:

Meaning and Concepts of Intellectual Property, Nature and Characteristics of Intellectual Property, Origin and Development of Intellectual Property, Kinds of Intellectual Property, Intellectual Property System in India, IPRs- Invention and Creativity- Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs)- **A brief summary of:** Patents, Copyrights, Trademarks, TRIPS and TRIMS , Industrial Designs- Integrated Circuits-Geographical Indications-Establishment of WIPO-Application and Procedures.

## Module-1

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### Research Meaning:

- Research in common man's language refers to "search for knowledge"
- Research is an art of scientific investigation.
- The Advanced Learner's Dictionary of Current English lays down the meaning of research as,

a careful investigation or inquiry specially through search for new facts in any branch of knowledge".

It is also a systematic design, collection, analysis and reporting the findings and solutions for problems of a company.

- Research is a structured enquiry that utilizes acceptable scientific methodology to solve problems and create new knowledge that is generally applicable.
- Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested.
- Research is an art of scientific investigation. It is also a systematic design, collection, analysis and reporting the findings and solutions for the problems of a company.

### Definition:

Research can be defined as an organized and systematic study of materials and sources in order to discover new things and establish facts and reach new conclusions. Essentially speaking, research involves a well-developed plan, a systematic approach to develop new theories or finding solutions to various problems.

According to Clifford Woody, “research comprises defining and refining problems, formulating hypothesis or suggested solutions, collecting, organizing and evaluating data, making decisions and reaching conclusions to determine whether they fit the formulating hypothesis”

### **Business Research**

- A Systematic inquiry that provides information to guide managerial decisions.
- Systematic and objective process of gathering, recording and analyzing data for aid in making business decisions.

### **Scientific Method**

Scientific research is one which yields the same results when repeated by different individuals. Techniques or procedures used to analyze empirical evidence in an attempt to confirm or disprove prior concepts

Empirical means verifiable by observation, experimentation or experience.

- A scientific undertaking which by means of logical and systematic techniques, aims to:
  1. Discover new facts or verify and test old facts.
  2. Analyze their sequences, interrelationships and causal explanations.
  3. Develop new scientific tools, concepts and theories, which would facilitate valid and reliable study of human behavior.

### **Characteristics of Scientific Method**

- **Validity** – Validity is the ability of a measuring instrument to measure what it is supposed to.
- **Reliability** – Measurement by instruments such as thermometer can be repeated any number of times by different individuals, but the results will be the same.

### **Types of research**

Research can be classified on the basis of purpose, time, methodology, setting etc. however one can broadly classify research as basic and applied research.

1. **Basic research:** research for the sake of enhancing knowledge is termed as basic research.
  - It is also called as theoretical research

- Basic or pure or fundamental research is done with the intention of empowering the unknown, Undertaken for the sake of knowledge without any intention to apply or practice.
- It is primarily concerned with developing and formulating theories and generalization.
- It addresses itself to more fundamental questions and not the problems with immediate commercial potential.
- Not directly involved with practical problems.
- Contributions:
  - i. By developing principle, pure research *offers solutions* to many *practical problems*.
  - ii. Helps to *find critical factors* in a practical problem.
  - iii. Develops many *alternative solutions*.
  - iv. How did the universe begin?
- What are protons, neutrons, and electrons composed of?
- What is the specific genetic code of the fruit fly?
- Example : Theory of Relativity (by Einstein)

## 2. Applied Research

- It is also termed as need based research having high practical relevance.
- Specific in nature, result oriented and is driven by clear aim.
- Undertaken to find solutions for real-life problem requiring an action or policy decision.
- It may incidentally contribute to the development of theoretical knowledge.
- Contributions:
  - Can contribute new facts about existing theories.
  - Puts theories to test.
  - May aid in conceptual clarification.

- Integrate previously existing theories. Example :
  - To develop a new market for the product.
  - improve the energy efficiency of homes, offices, or modes of transportation
- 3. **Qualitative Research** – An unstructured, exploratory research methodology based on small samples that provides insights and understanding of the problem setting.
  - The aim is a complete, detailed description.
  - Data is in the form of words, pictures or objects.
  - Case studies, group interviews
- 4. **Quantitative Research** – A research methodology that seeks to quantify the data and typically applies some form of statistical analysis.
  - The aim is to classify features, count them, and construct statistical models in an attempt to explain what is observed.
  - Data is in the form of numbers and statistics.
  - Experiments and surveys
- 5. **Historical Research**
  - Historical study is a study of past records and data in order to understand the future trends and development of the organization or market.
  - There is no direct observation.
  - Research has to depend on the conclusions or inferences drawn in the past.
  - Main objective is to derive explanation and generalization from the past trends in order to understand the present and anticipate the future.
  - Example : Investors in the share market study the past records or prices of shares which he/she intends to buy. Studying the share prices of a particular company enables the investor to take decision whether to invest in the shares of a company.
- 6. **Exploratory Research:** this kind of research is carried out at the very beginning when the problem is not clear or vague.

- Provides insights into and comprehension of an issue or situation.
- Often relies on secondary research such as reviewing available literature and/or data, or qualitative approaches such as informal discussions with consumers, employees, management or competitors and more formal approaches through in-depth interviews, focus groups, case studies or pilot studies.
- Example : Exploring the reasons for sales decline of a product in a company.

## 7. Descriptive Research

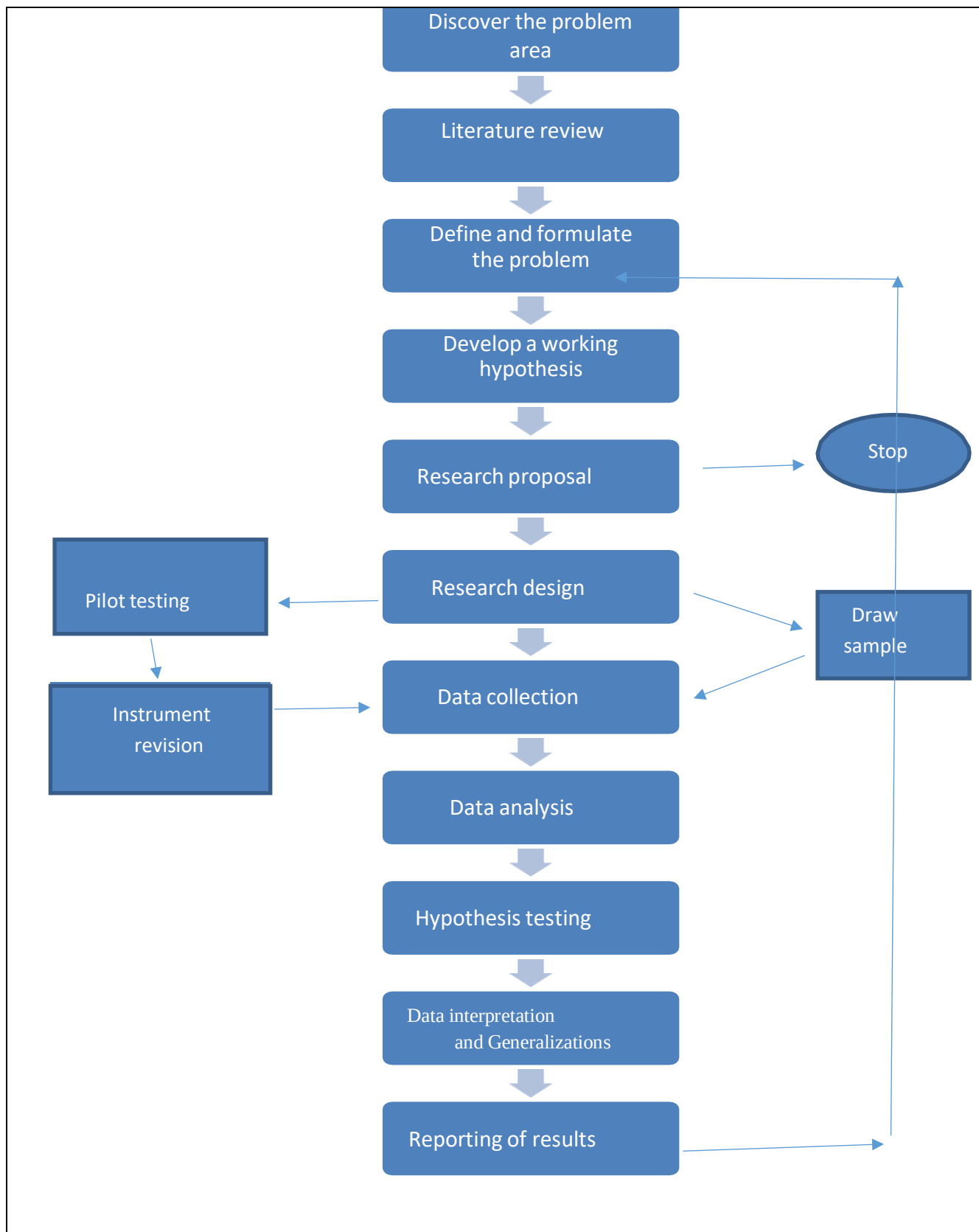
- Describes data and characteristics about the population or phenomenon being studied.
- Answers the questions who, what, where, when and how.
- Describes the characteristics of the respondent in relation to a particular product. Deals with demographic characteristics of the consumer.
- Example : Degree of viewing TV Channels, its variation with age, income level, profession of respondent as well as time of viewing.

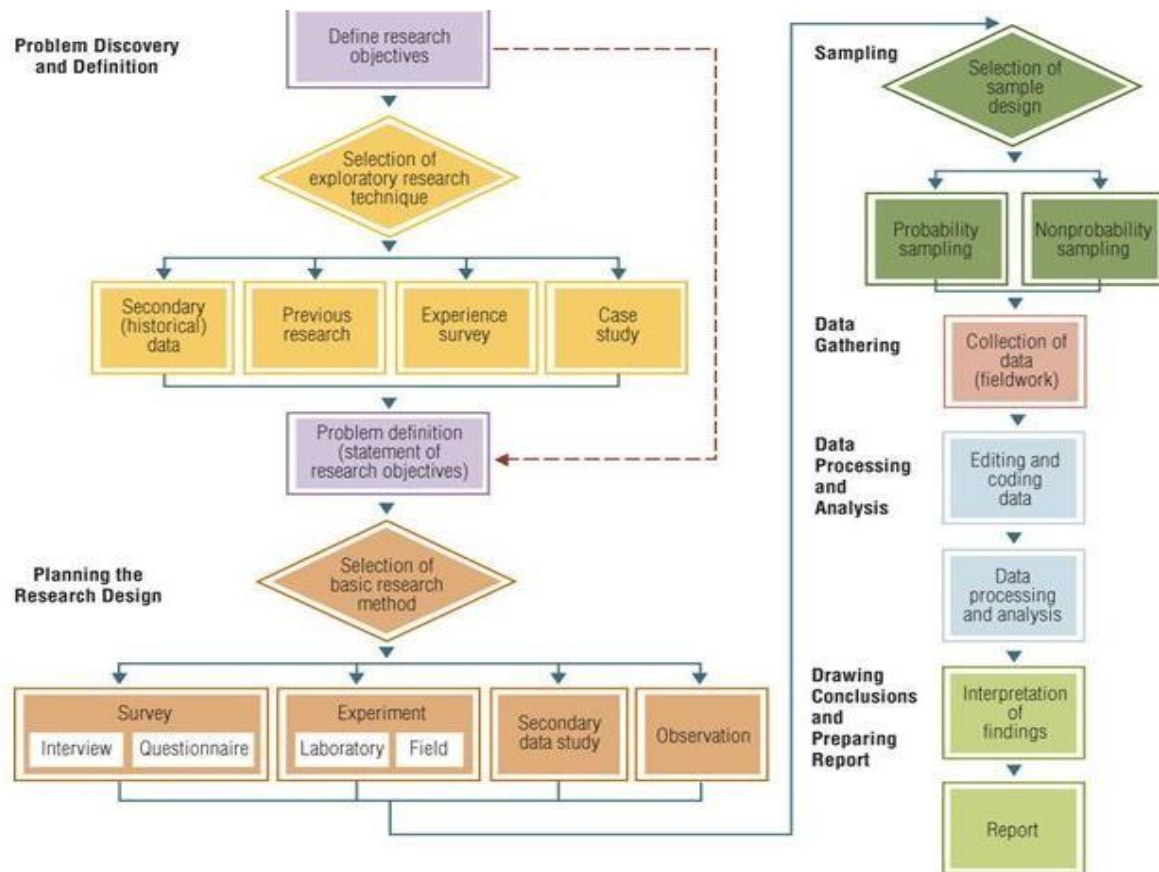
## 8. Causal Research

- Conducted to determine the cause and effect relationship between the two variables.
- Way of seeing how actions now will affect a business in the future.
- Example:
  - Training and productivity of an employees
  - Advertisement and sales

## 9. Simulation research

- This research involves the creation of an artificial environment which is very similar to the real environment. Therefore with in the artificial environment the variables are manipulated and studied.
- Ex: study of consumer buying behavior by giving specific amount of time and money in a mall.

**Research process**



Note: Diamond-shaped boxes indicate stages in the research process in which a choice of one or more techniques must be made. The dotted line indicates an alternative path that skips exploratory research.

1. **DISCOVERING THE PROBLEM AREA:** to start with a research, one needs to discover the problem which needs solution. Any researcher is pretty sure of the subject on which to conduct his research, but the exact problem area may remain exclusive.

The best way to identify the problem would be to look for an unresolved query, a gap in the existing knowledge or an unfulfilled need within the chosen subject. The researcher should take care that the problem should be the one which can be clearly defined and formulated. Further while choosing the research area, the researcher should look into the availability of information relevant to the topic. Mere availability is not enough. It must be accessible.

2. **LITERATURE REVIEW:** it involves a comprehensive review of published and unpublished work from the secondary sources of data available in the relevant area of study.

Literature review helps the researcher in 2 ways

- Specifying his research in a meaningful context
- It would provide him with an insight into the methods and techniques adopted for handling such problems

3. **PROBLEM DEFINITION:** initially the focus of the problem is ambiguously defined. However after the literature review, the researcher is in a position to formulate problem clearly. A problem well formulated alone yield fruitful results. It is a clear, precise and concise statement of the matter that is to be investigated.

4. **DEVELOPING A WORKING HYPOTHESIS:** hypothesis is a tentative hypothesis regarding a solution to the problem under study.

Hypothesis is a predictive statement which is made in the light of available facts relating to the problem under study.

The type of data to be collected and the tools of analysis are influenced by the hypothesis.  
Ex: a study conducted to find out the amount of research investment being done by the companies will have the following hypothesis.

H0: companies invest 1% of their sales revenue in research activities (null hypothesis)

H1: companies invest less than 1% of their sales revenue in research activities (alternative hypothesis)

5. **RESEARCH PROPOSAL:** it is a brief summary outlining the objectives of study and the modus operandi of conducting the research.

In case of thesis, the research proposal are in the form of synopsis stating

- Research objectives
- Proposed methodology of research
- Benefits of study
- Detailed bibliography

In case of government or business organizations, in addition to the above information there are other details like

- Researcher's qualification
- Time and cost schedule
- Resources and special facilities required during the research.

Essentially they contain an executive summary in the beginning stating the objectives of study and benefits derived thereon.

6. **RESEARCH DESIGN:** it outlines the conditions for collection and analysis of data. The what, when, where, how much and the method of data collection are detailed in the research

design. It will specially contain information about:

- **Sampling design**
- **Data collection design**
- **Instrument design**

**Sampling design:** the plan devised to draw a sample is termed as sampling design. A sampling design will include a decision on the

- Sampling unit
- Sampling technique
- Sample size

**Data collection design:** the method of gathering data is planned in data collection design. Briefly speaking, some of the data collection techniques available to researcher are

- Questionnaire
- Interview
- Observation

**Instrument design:** instrument designing schedule or the questionnaire the researcher would use to collect data. The researcher while designing the questionnaire must think in terms of

- Type of data
- Communication approach
- Question structure
- Question wording etc

## 7. DATA COLLECTION

This involves sending questionnaire to respondents, training field workers in interview and observation methods and making them to collect necessary data.

## 8. DATA ANALYSIS

All the data gathered in the previous step has no meaning till the time it is not subject to data analysis.

Processing of raw data will yield some kind of relevant information. Data analysis is concerned with reducing the bulk of accumulated data to a manageable size.

Generally speaking, it involves the following procedures:

- Editing

- Coding
- Tabulation
- Statistical analysis

## 9. HYPOTHESIS TESTING

There are various parametric and non-parametric tests like z-test, t-test, f-test etc

Hypothesis testing helps a researcher to establish if the difference between values is due to chance or actual.

## 10. DATA INTERPRETATION AND GENERALISATION(finding out meaning)

Researcher can use inductive or deductive approach. After the data is been analysed and tested repeatedly for arriving at conclusive results, generalizations are made to build new theory.

## 11. REPORTING OF RESULTS

The last step is concerned with bringing in public the results of the research so that the design can be put to some use.

### **Format of the report**

#### **Preliminary section**

Title of the report, table of contents, list of tables, graphs, preface and an executive summary

#### **Main text**

Problem background, research objectives, research methodology, importance of conducting research and the conclusion arrived thereof.

#### **End section**

Appendices supporting the research items like questionnaire, schedule used etc and bibliography.

### **Management problem**

When a problem has been clearly and accurately identified can a research project be conducted properly. It is a concise description of the issues that need to be addressed by a problem solving team and should be presented to them (or created by them) before they try to solve the problem. It asks what the decision maker needs to do. Mostly action oriented and Focuses on symptoms.

### **Defining the research problem**

The research problem is a general statement of an issue meriting research. Its nature will suggest appropriate forms for its investigation

Problem definition involves stating the general marketing research problem and identifying its specific components. Only when research problem has been clearly defined can research be designed and conducted properly.

### Management Problem vs Research Problem

**Management Problem** involves “what needs to be done?”

**Research Problem** involves “what information needs to be provided and how can the information is obtained”. It refers to some difficulty an organization faces and wishes to obtain solution for the same.

| Basis of comparison  | Management problem                         | Research problem                                                                                                       |
|----------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Nature</b>        | Asks what the Decision Makers needs to do  | Asks what information needed and how it can be obtained                                                                |
| <b>Orientation</b>   | Action Oriented                            | Information oriented                                                                                                   |
| <b>Focus</b>         | Focuses on Symptoms                        | Focuses on underlying causes                                                                                           |
| <b>Common errors</b> | Problem definition is too broad            | Problem definition is too narrow                                                                                       |
| <b>example</b>       | Should a new product be introduced         | To determine consumer preferences and purchase intentions for the proposed new product                                 |
| <b>example</b>       | Should the advertising campaign be changed | To determine the effectiveness of the current advertising campaign                                                     |
| <b>example</b>       | Should the price of the brand be increased | To determine the price elasticity of the demand and the impact on sales and profits of various levels of price changes |

### **Formulating the research Hypothesis**

A good hypothesis relates and explains the known facts. It should also predict new facts. It *must* be stated in such a way that we can test it by experimentation or further observation, or it is of no scientific value. Also, it must be stated in a way that would allow us to show if it is incorrect, i.e., it must be "falsifiable." A scientist *must* be willing to accept the possibility that his or her hypothesis is incorrect, and this point often separates true science from pseudoscience. (In fact, most scientists work hard to develop good hypotheses, and then spend a great deal of effort trying to *disprove* them. Pseudo scientists tend to settle on a hypothesis that suits their needs or expectations, and then spend a great deal of effort trying to prove that they are "*true*".

### **Developing the research proposals**

Research proposal is outline, a draft plan of a research work proposed by the researcher. It is also termed as synopsis. It contains

- A statement of the problem
- The proposed methodology
- The benefits of the research
- The reasons required to conduct the research

### **Contents of research proposal**

The research proposal contents differ depending on the type of proposal and intended audience.

1. Problem statement
2. Research objectives
3. Literature review
4. Importance/ benefits of study
5. Research design
6. Data analysis
7. Nature and form of results
8. Schedule
9. Facilities and special resources
10. Bibliography
11. Appendices
12. Measurement instrument

### **Research design formulation**

A research design is a framework or blueprint for conducting the marketing research project. It details the procedures necessary for obtaining the required information, and its purpose is to design a study that will test the hypotheses of interest, determine possible answers to the research questions, and provide the information needed for decision making. Decisions are also made regarding what data should be obtained from the respondents (e.g., by conducting a survey or an experiment). A questionnaire and sampling plan also are designed in order to select the most appropriate respondents for the study. The following steps are involved in formulating a research design:

- Secondary data analysis (based on secondary research)
- Qualitative research
- Methods of collecting quantitative data (survey, observation, and experimentation)
- Definition of the information needed
- Measurement and scaling procedures
- Questionnaire design
- Sampling process and sample size
- Plan of data analysis

### **Sampling design**

Sampling is a means of selecting a subset of units from a target population for the purpose of collecting information. This information is used to draw inferences about the population as a whole. The subset of units that are selected is called a sample. The sample design encompasses all aspects of how to group units on the frame, determine the sample size, allocate the sample to the various classifications of frame units, and finally, select the sample. Choices in sample design are influenced by many factors, including the desired level of precision and detail of the information to be produced, the availability of appropriate sampling frames, the availability of suitable auxiliary

variables for stratification and sample election, the estimation methods that will be used and the available budget in terms of time and resources.

### **Planning and collecting the data for research**

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes. Generally there are three types of data collection and they are

1. Surveys: Standardized paper-and-pencil or phone questionnaires that ask predetermined questions.

2. Interviews: Structured or unstructured one-on-one directed conversations with key individuals or leaders in a community.

3. Focus groups: Structured interviews with small groups of like individuals using standardized questions, follow-up questions, and exploration of other topics that arise to better understand participants

### **Data analysis and interpretation**

Analysis of data is a process of inspecting, cleaning, transforming, and modeling data with the goal of discovering useful information, suggesting conclusions, and supporting decision-making. Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, in different business, science, and social science domains.

### **Literature review**

It involves a comprehensive review of published and unpublished work from the secondary sources of data available in the relevant area of study.

Importance of literature review:

1. Identification of problem for the study
2. Monitors the repetition
3. Helps avoiding mistakes
4. Creates appropriate theoretical structure
5. Facilitates interpretation of study

## Research application in business decision

☐ Businesses use research to ascertain the success of their advertising. For instance, a dairy manufacturer may want to find out what percentage of the people saw its latest TV commercial. The dairy manufacturer may find that more individuals become aware of its advertising the longer the TV ad runs. The company may decide to run its TV ads at different times if few people have seen the ads.

☐ A business can make well-informed decisions due to research. In the research process, the business will be able to acquire details about key business areas, analyze it, create a strategy and distribute business information. Reports, presented to the top management, often contain details on consumer and employee preferences and all the available channels for sales, marketing, finance and production. Management makes use of these details to **determine the best strategy**.

☐ Research is a necessity at all stages and phases of business operations. Initial research is needed to evaluate whether getting into the given kind of business would be profitable and whether there exists demand for the proposed product.

☐ Regarding the staff, a correctly carried out research can uncover important details on their satisfaction quotient, the difficulties experienced by them and how the problems related to relationships at the workplace could be handled. An analysis into the results would allow the management to bring about changes for the all round effective functioning of the organization and its employees. The workers **can be trained and coached in line with the needs**. This would help personal as well as professional development improving overall organizational performance.

☐ Research is important for **managerial decision making**. All strategic business areas are analyzed and evaluated; then techniques for more efficient procedures are created. Through proper research, the organization will be able to pick the most effective, productive and profitable one.

☐ Research could possibly be applied to marketing, production, finance, IT and Human resources.

☐ Research can answer questions for **various problems**, from getting a grip on industry trends, identifying new products to produce and deliver to the market, or deciding on which site to locate an outlet, to better understanding what it needs to fulfill customer demands. Research can also help evaluate if a product is accepted in the market. Research aids expansion into new markets.

☐ Research helps in **testing the potential success of new products**. Businesses have to understand what kinds of products consumers would like before they market them. For instance, a restaurant may in the beginning, interview focus groups to test types of burgers. The focus groups will probably include small teams of consumers. The goal of the focus group may be to figure out which burger customers prefer. Ultimately, the company may test the burgers through surveys with larger groups of people. The above points state the importance of research in business decision making. Research is necessary to gather facts and statistics with regards to a company's customers, employees and competitors. Based on these numbers, businesses are able to make better managerial decisions.

☐ **New Knowledge:** The fascination and desire for new knowledge, new facts for business cycles, environment analysis and technological up gradation are the primary reasons of research.

☐ **Research assists** the company to find the right supplier at the right price and at the right time. An appropriate supplier choice makes it possible for the company to obtain or acquire top quality raw materials which result into production of good quality items which are consumed by the end user.

## **Ethical issues in business research.**

### **1 . Study design and ethics approval**

According to COPE, “good research should be well adjusted, well-planned, appropriately designed, and ethically approved. To conduct research to a lower standard may constitute misconduct.” This may appear to be a stringent criterion, but it highlights the basic requirement of a researcher is to conduct a

research responsibly. To achieve this, a research protocol should be developed and adhered to. It must be carefully agreed to by all contributors and collaborators, and the precise roles of each team member should be spelled out early, including matters of authorship and publications. Research should seek to answer specific questions, rather than just collect data.

It is essential to obtain approval from the Institutional Review Board, or Ethics Committee, of the respective organisations for studies involving people, medical records, and anonymised human tissues. The research proposal should discuss potential ethical issues pertaining to the research. The researchers should pay special attention to vulnerable subjects to avoid breach of ethical codes (e.g. children, prisoners, pregnant women, mentally challenged, educationally and economically disadvantaged). Patient information sheet should be given to the subjects during recruitment, detailing the objectives, procedures, potential benefits and harms, as well as rights to refuse participation in the research. Consent should be explained and obtained from the subjects or guardians, and steps should be taken to ensure confidentiality of information provided by the subjects.

## **2. Data analysis**

It is the responsibility of the researcher to analyse the data appropriately. Although inappropriate analysis does not necessarily amount to misconduct, intentional omission of result may cause misinterpretation and mislead the readers. Fabrication and falsification of data do constitute misconduct. For example, in a clinical trial, if a drug is found to be ineffective, this study should be reported. There is a tendency for the researchers to under-report negative research findings, and this is partly contributed by pressure from the pharmaceutical industry which funds the clinical trial.

To ensure appropriate data analysis, all sources and methods used to obtain and analyse data should be fully disclosed. Failure to do so may lead the readers to misinterpret the results without considering possibility of the study being underpowered. The discussion section of a paper should mention any issues of bias, and explain how they have been dealt with in the design and interpretation of the study.

## **3. Authorship**

There is no universally agreed definition of authorship. It is generally agreed that an author should have made substantial contribution to the intellectual content, including conceptualising and designing the study; acquiring, analysing and interpreting the data. The author should also take responsibility to certify that the manuscript represents valid work and take public responsibility for the work. Finally, an author is usually involved in drafting or revising the manuscript, as well as approving the submitted manuscript. Data collection, editing of grammar and language, and other routine works by itself, do not deserve an authorship.

It is crucial to decide early on in the planning of a research who will be credited as authors, as contributors, and who will be acknowledged. It is also advisable to read carefully the “Advice to Authors” of the target journal which may serve as a guide to the issue of authorship.

#### **4. Conflicts of interest**

This happens when researchers have interests that are not fully apparent and that may influence their judgments on what is published. These conflicts include personal, commercial, political, academic or financial interest. Financial interests may include employment, research funding, stock or share ownership, payment for lecture or travel, consultancies and company support for staff. This issue is especially pertinent in biomedical research where a substantial number of clinical trials are funded by pharmaceutical company.

Such interests, where relevant, should be discussed in the early stage of research. The researchers need to take extra effort to ensure that their conflicts of interest do not influence the methodology and outcome of the research. It would be useful to consult an independent researcher, or Ethics Committee, on this issue if in doubt. When publishing, these conflicts of interest should be declared to editors, and readers will judge for themselves whether the research findings are trustworthy.

#### **5. Redundant publication and plagiarism**

Redundant publication occurs when two or more papers, without full cross reference, share the same hypothesis, data, discussion points, or conclusions. However, previous publication of an abstract during the proceedings of meetings does not preclude subsequent submission for publication, but full disclosure should be made at the time of submission. This is also known as self-plagiarism. In the increasing competitive environment where appointments, promotions and grant applications are strongly influenced by publication record, researchers are under intense pressure to publish, and a growing minority is seeking to bump up their CV through dishonest means.

On the other hand, plagiarism ranges from unreferenced use of others’ published and unpublished ideas, including research grant applications to submission under “new” authorship of a complete paper, sometimes in different language.

Therefore, it is important to disclose all sources of information, and if large amount of other people’s written or illustrative materials is to be used, permission must be sought.

#### **Features of a good research study.**

- **Systematic-** research should be structured.
- **Logical-** there must be logical reasoning
- **Empirical-** factual investigation must be possible. Its validity can be checked through

reliable sources of evidences.

- **Replicable-** the research conducted can be repeated any number of times and yield the same result
- **Purpose clearly defined** – The purpose of the business research-the problem involved or the decision to be made-should be clearly defined and sharply delineated in terms as unambiguous as possible. The statement of the decision problem should include its scope, its limitations, and the precise meanings of all words and terms significant to the research.
- **Research process detailed** – The research procedures used should be described in sufficient detail to permit another researcher to repeat the research. Except when secrecy is imposed, research reports should reveal the sources of data and the means by which they were obtained.
- **Research design thoroughly planned** – The procedural design of the research should be carefully planned to yield results that are as objective as possible. A survey of opinions or recollections ought not to be used when more reliable evidence is available from documentary sources or by direct observation.
- **High ethical standards applied** – Ethical issues in research reflect important moral concerns about the practice of responsible behavior in society. A research design that includes safeguards against causing mental or physical harm to participants and makes data integrity a first priority should be highly valued.
- **Limitations frankly revealed** – The researcher should report, with complete frankness, flaws in procedural design and estimate their effect on the findings.
- **Analysis adequate for decision maker's needs-** Analysis of the data should be extensive enough to reveal its significance, what managers call “insights”. The methods of analysis used should be appropriate
- **Findings presented unambiguously** – Language that is restrained, clear, and precise; assertions that are carefully drawn and hedged with appropriate reservations; and an apparent effort to achieve maximum objectivity tend to leave a favorable impression of the researcher with the decision maker.
- **Conclusions justified** – Conclusions should be limited to those for which the data provide an adequate basis. Good researchers always specify the conditions under which their conclusions seem to be valid.

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